Marimuthu Mayavan

Seeking a 6-month internship from February 2027 – IoT, Networks, Mobile Security, Embedded Systems

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EDUCATION

M.Sc. (Diplôme National de Master), Networks and Telecommunications – IoT track Sep 2025 – 2027 (expected)
EURECOM, Sophia Antipolis, France

- Coursework: IoT Protocols, Network Security, Mobile and Smartphone Security, Deep Learning

Integrated M.Tech., Computer Science and Engineering (6-year programme) 2017 – 2023
Bharathidasan University, Tiruchirappalli, India · Coursework: Computer Networks, Network Security, IoT, Machine Learning

- Final project: built a bank CCTV theft-detection application (C#/.NET, SQL Server) that detects motion in restricted hours, captures faces (Haar cascade) and sends email and SMS alerts with the face image
- Project (2021): detected crop rows and failures in paddy fields from drone (UAV) images using wavelet-based clustering and morphological segmentation in MATLAB

PROFESSIONAL EXPERIENCE

L1 Network Engineer, IT Department Dec 2023 – Nov 2024
Orbion Pharmaceuticals (via Groveman Global, apprenticeship), Chennai area, India

- Provided first-level network support for 80+ PCs: LAN and Wi-Fi connectivity, IP configuration, switches and access points
- Diagnosed user connection problems with standard tools (ping, traceroute, ipconfig) and escalated complex issues to L2
- Installed and configured desktops, printers and network devices for employees

Trainee, Security Department (CCTV Operator) Apr 2025 – Aug 2025
Foxconn Hon Hai Technology India, Chennai, India · monitored a networked CCTV system across a large manufacturing site

RESEARCH EXPERIENCE

Modelling Automotive Systems Using Ansys System Architecture Modeler Sep 2026 – present
EURECOM research project (200 hours) with Ansys, supervised by Rabea Ameur-Boulifa · SysML v2, MBSE

- Modelling automotive system architectures in SysML v2 with Ansys System Architecture Modeler, using model-based systems engineering (MBSE) on automotive use cases provided by Ansys

Designing an Engaging Course on Audiovisual Deepfake Detection Feb 2026 – Jun 2026
EURECOM research project (100 hours), supervised by Prof. Jean-Luc Dugelay

- Reviewed recent research (IEEE Xplore, ACM, arXiv) on audio-visual feature fusion, lip-sync-based detection, datasets (FakeAVCeleb, DFDC, LAV-DF) and metrics (AUC-ROC, EER)
- Built 5 course modules with quizzes and a narrated AI-generated video lecture (HeyGen) for the course section on audiovisual deepfake detection

PROJECTS

Human Activity Recognition from Wearable IoT Sensors · Python, PyTorch, CNN/LSTM Apr 2026
EURECOM Deep Learning project · github.com/Mayavan-m/HAR-DeepLearning

- Built and compared 1D-CNN, LSTM and CNN-LSTM models on accelerometer and gyroscope data from 30 people (UCI HAR dataset)
- Reached 94% accuracy with the LSTM (94% on unseen people); showed the system still works without the gyroscope (93% → 82%)

PickErUp – Autonomous Mining Robot (LEGO EV3) · C, Linux (ev3dev), Git Dec 2025 – Jan 2026
EURECOM Operating Systems project, team of 3 · gitlab.eurecom.fr/os_group3/the_picker

- Wrote the robot's basic movement and rotation functions in C (motor control) for an autonomous robot that finds ores, picks them up with a claw and brings them back to its box
- Helped integrate the main control program; robot used sonar, colour and gyro sensors with 4 motors on an EV3 brick running Debian Linux

Privacy Impact Assessment of PayPal "Pay in 4" (AI credit scoring) · GDPR, CNIL PIA, AI Act Oct 2026 – ongoing
EURECOM Security and Privacy for Big Data and Cloud course, team of 3

- Analysing PayPal's privacy policy, assessing privacy risks with the CNIL PIA tool (acting as DPO) and classifying the automated creditworthiness system under the EU AI Act

SKILLS, CERTIFICATION AND LANGUAGES

Programming & ML: Python, C, C++, Java, MATLAB · PyTorch, CNN, LSTM, model evaluation
Networking & IoT: TCP/IP, LAN/Wi-Fi troubleshooting, network security basics · MQTT, CoAP, wearable sensor data
Modelling & tools: SysML v2, UML, MBSE, Ansys System Architecture Modeler · Linux, Windows, Git/GitHub, Jupyter
Certification: Diploma in Computer Programming (C, C++, Python), TCDS Computer Education, 2023 – Distinction (91/100)
Languages: English (fluent), Tamil (native), French (A1)